

Meaningful Human Control

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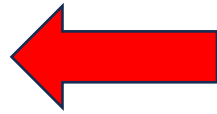
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Overview of the course so far

- Basic sources of ethical premises/ ethical argumentation
- AI Terminology and Philosophical Background
- Design for Values
- Values and AI
 - Responsibility
 - Fairness
 - Sustainability
 - Explicability/ Explainability
- Automation of Human Tasks
 - Meaningful human control
 - Building ethics into machines
 - Risks of superintelligence

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Rule-based expert systems, example 1

MYCIN Rule 507

IF:

1. The infection requiring therapy is meningitis
2. Organisms were not seen on the stain of the culture
3. The type of infection is bacterial
4. The patient does not have a head injury defect, AND
5. The age of the patient is between 15 years and 55 years

THEN: The organisms that might be causing the infection are

Diplococcus-pneumoniae and
Neisseria-meningitidis

Musen MA, Middleton B, Greenes RA. Clinical Decision-Support Systems. Biomedical informatics: 643-74. 2014.

Rule-based expert systems, example 2

Oncoguide

Oncoguide biedt zorgprofessionals inzichten uit richtlijnen, de Nederlandse Kankerregistratie (NKR) en predictiemodellen.

Op basis van patiënt- en ziektekenmerken toont Oncoguide stap voor stap een route door beslisbomen naar zorgaanbevelingen op basis van richtlijnen. Oncoguide toont informatie uit verschillende bronnen op een transparante manier in beslisbomen op basis van de medische richtlijn.

[Oncoguide \(iknl.nl\)](http://iknl.nl)

Advantages mentioned by oncologists:

- 1) It is far more usable than a text document.
- 2) It allows scientific developments to be immediately introduced into the decision tree.
- 3) It can be used directly in patient consultations.
- 4) Deviations from the guidelines and grey area judgments can be explained to and decided upon with patients with the decision tree as a background.
- 5) It can be used in medical training.
- 6) It could be used for marking patients as suitable to recruit for a given cancer study, based on EHR data.

Meaningful Human Control

Background: why people care about control and responsibility

- One of the central concerns about a (semi-)autonomous system interacting with humans is that nobody can be reasonably held responsible for bad outcomes caused by the system. This has been called a ***responsibility gap***.
- In order to be held responsible, an entity must satisfy the following conditions:
 - Foreknowledge (“knew or should have known”)
 - Control/ causal influence
 - Freedom (“could have done otherwise”)

“Meaningful human control”

“humans not computers and their algorithms should ultimately remain in control of, and thus morally responsible for, relevant decisions” (Santoni di Sio & van den Hoven 2018).

This is more than merely a “human in the loop”.

“MHC is still an open concept that needs interpretation” (Hille, Hummel, & Braun 2023).

Meaningful Human Control over AI for Health? A Review

Eva Maria Hille ¹, Patrik Hummel ², Matthias Braun ¹

ABSTRACT

Artificial intelligence is currently changing many areas of society. Especially in health, where critical decisions are made, questions of control must be renegotiated: who is in control when an automated system makes clinically relevant decisions? Increasingly, the concept of meaningful human control (MHC) is being invoked for this purpose.

of, for example, potentially imperfect human cognitive capacity to grasp and respond to complex situations. Second, the limited capacity and behaviour. And third, the limited capacity of causal control might not be enough to ensure the kinds of moral or legal responsibility that require stricter control conditions. In health, there are various additional considerations.

The (im)possibility of meaningful human control for lethal autonomous weapon systems

August 29, 2018, Analysis / Autonomous Weapons / Law and Conflict / Weapons
13 mins read

AI and Ethics

<https://doi.org/10.1007/s43681-023-00300-0>

ORIGINAL RESEARCH



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The many meanings of meaningful human control

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Abstract

The concept of Meaningful Human Control (MHC) has gained prominence in the field of artificial intelligence. It is discussed in relation to lethal autonomous weapons, autonomous cars, and mor

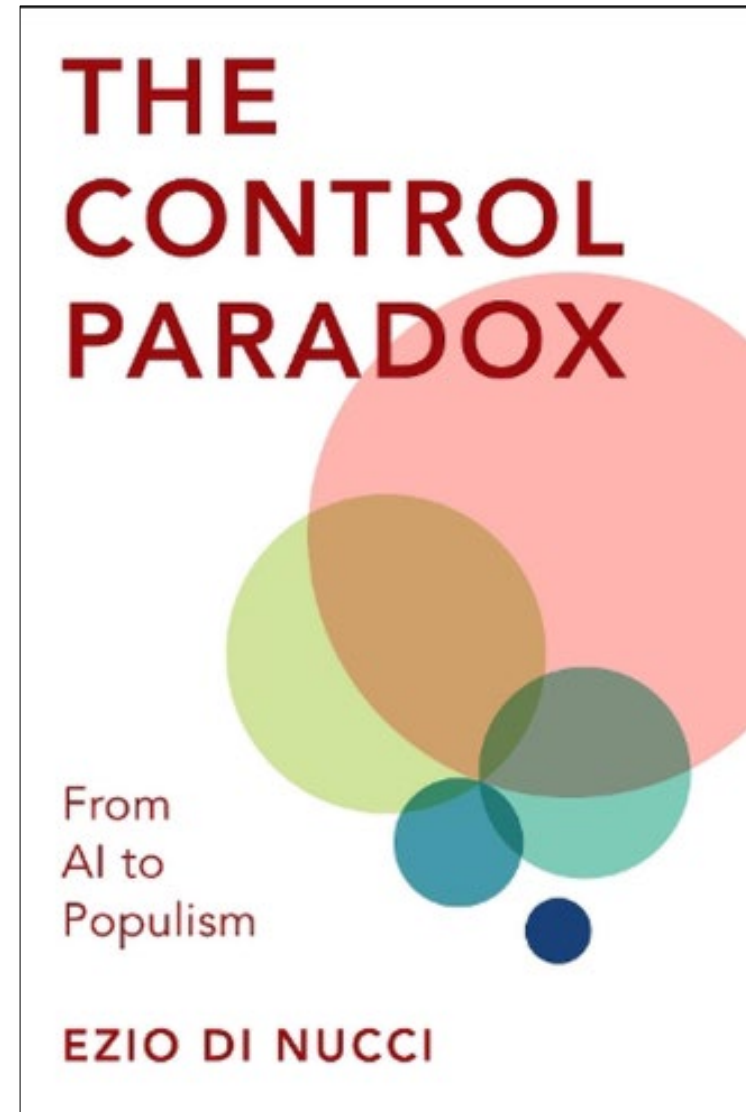
Concepts of control

Guidance control

Regulatory control

Being in control vs. controlling

Relationship of responsibility and control



MHC

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THE CONTROL PARADOX

From
AI to
Populism

EZIO DI NUCCI

Two dimensions of control (Fischer & Ravizza)

Guidance control (the basis of “meaningful human control” in Santoni di Sio & van den Hoven 2018).

- Reason-responsiveness
- Each outcome is attributable to an agent

Regulative control

- The agent could have acted differently (freedom)

Two dimensions of control (Fischer & Ravizza)

MHC

Guidance control (the basis of “meaningful human control” in Santoni di Sio & van den Hoven 2018).

- Reason-responsiveness (tracking in MHC)
- Each outcome is attributable to an agent (tracing in MHC)

Regulative control

- The agent could have acted differently (freedom --- *not* part of MHC)

Meaningful human control (Santoni de Sio & van den Hoven 2019)

- Meaningful human control means guidance control.
- Regulative control (deeper freedom of action) is not required.
- Meaningful human control has two aspects: tracking and tracing. These are meant to “unpack” the idea of guidance control (and thus of MHC as a whole).

Aims of MHC

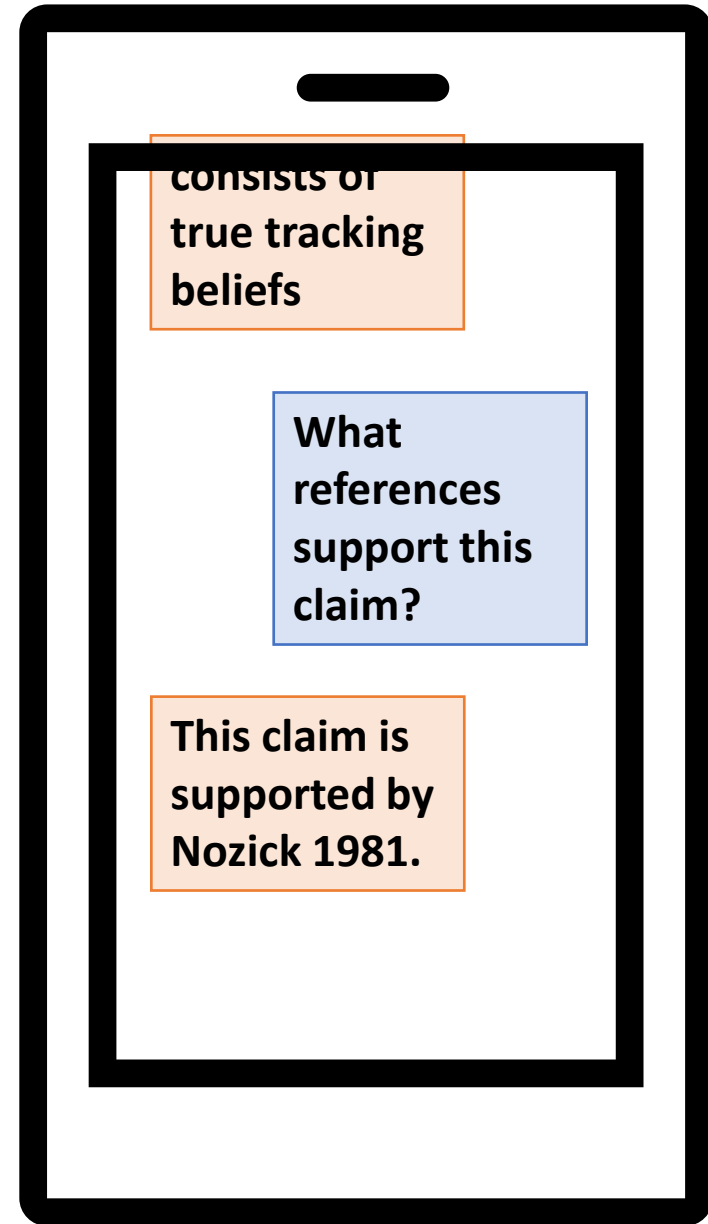
- “Delegation of activity to smart machines is compatible with genuine responsibility attributions, even if the machine’s decisions are not free ” (Santoni de Sio & van den Hoven, p. 5).
- Humans can be responsible: “there is a difference between human actions and other natural events, and they thus claim that human agents can be legitimately seen as morally responsible for at least some of their actions” (ibid.).
- The main idea is that when guidance control is satisfied by automation, the humans who built and operate the system can be held responsible; and when it isn’t satisfied, something has gone wrong that should have been prevented during the design process.

Tracking condition in epistemology (Nozick 1981)

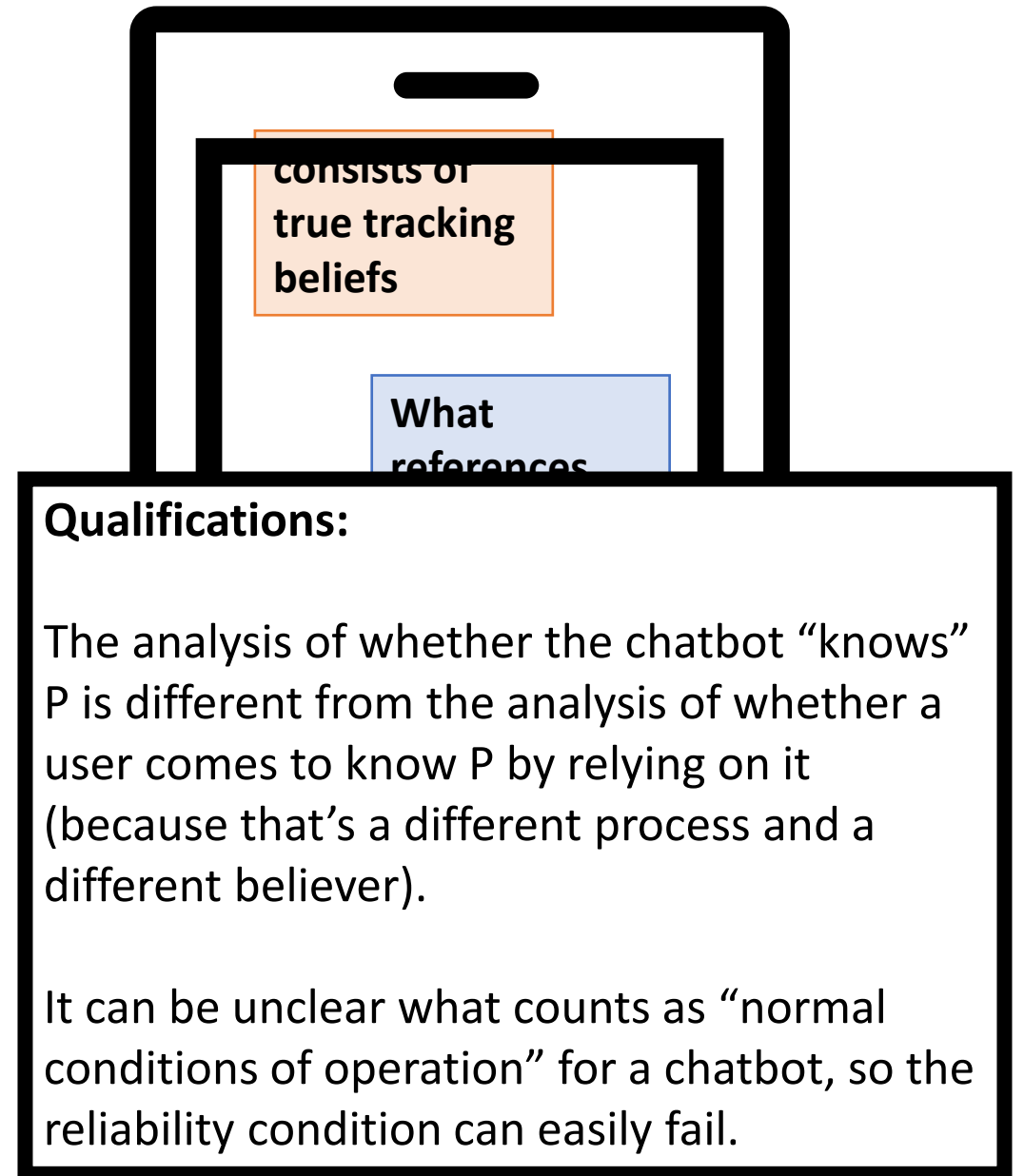
The tracking condition comes from the theory of knowledge. Knowledge of S that B is analyzed by Nozick as consisting of the following components:

- (i) P is true (Federer is the Wimbledon champion).
- (ii) S believes that P (I believe that Federer is the Wimbledon champion).
- (iii) If it were not the case that P, then S would not believe that P (if Federer was not the Wimbledon champion, I would not believe that he is the Wimbledon champion).
- (iv) If it were the case that P, then S would believe that P (if Federer was the Wimbledon's champion, then I would believe that he is the Wimbledon's champion).

Tracking condition for AI chatbots



Tracking condition for AI chatbots



What does tracking mean for a drone?



Drone epistemology

When we say that a drone satisfies the tracking condition for some true propositions $p_1 - p_n$, if and only if:

- The drone ***represents*** propositions $p_1 - p_n$ as true, sufficient to guide the domain of actions over which the drone operates according to relevant norms, under predictable operating conditions.
- The drone would not represent $p_1 - p_n$ as true if they were not true.
(Safety)
- The drone would represent $p_1 - p_n$ as true if they were true.
(Sensitivity)

What does tracing mean for a drone?



Automation and responsibility

automation

full automation

human in the loop

human must
override system in
some conditions

system must
override human in
some conditions



<https://www.youtube.com/watch?v=f0P1lkyz8To>
<https://www.cnet.com/pictures/the-increasingly-autonomous-robots-of-war-pictures/>

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Tracing condition

Background idea: “ownership” of an action or decision (Fischer & Ravizza); being able to attribute actions or decisions to specific humans or organizations.

- The drone cannot own its own actions. Hence the tracing condition is about other agents: the deployer, the designer, and the operator.

Some general principles under the tracing condition

- When the drone acts (non-accidentally) correctly by design under standard conditions, then its actions can be attributed to the deployer and the designer.
- When it fails under standard conditions, then its actions can be attributed to the designer.
- When it fails under non-standard conditions, then its actions can be attributed to the deployer, the designer, or the operator.

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Some principles under the tracing condition for human-in-the-loop cases

- If by the design the operator was reasonably expected to override the drone but did not, then its actions can be attributed to the operator.
- If by the design the operator was not expected to override the drone and they did not, then its actions can be attributed to the deployer.
- If by the design the operator was unreasonably expected to override the drone and they did not, then its actions can be attributed to the designer.

Responsibility and design

“many of the choices made by designers can be seen as decisions about what should be delegated to a machine and what should be left to the initiative of human actors.”

Akrich 1992, P. 216

Responsibility and design

“many of the choices made by designers can be seen as decisions about what should be delegated to a machine and what should be left to the initiative of human actors.”

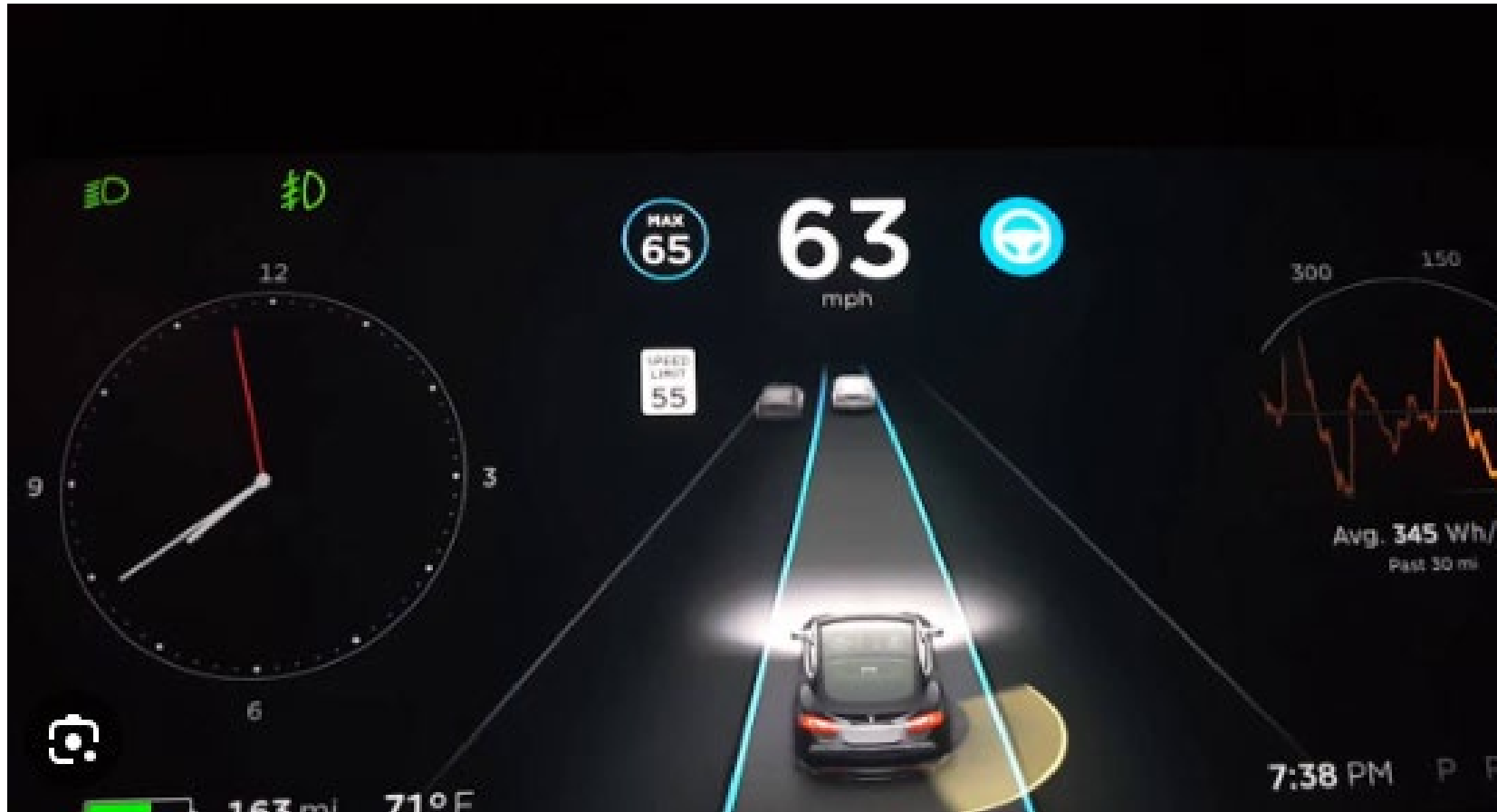
Akrich 1992, P. 216



Image:
<https://industrialscripts.com/stage-directions/>

Implications for designers

“Designing for satisfying the tracing condition means ensuring that different human agents along the chain are technically and psychologically capable of complying with their tasks and are well aware of their responsibility for the behavior of the autonomous system.” (Santoni de Sio & van den Hoven 2018).

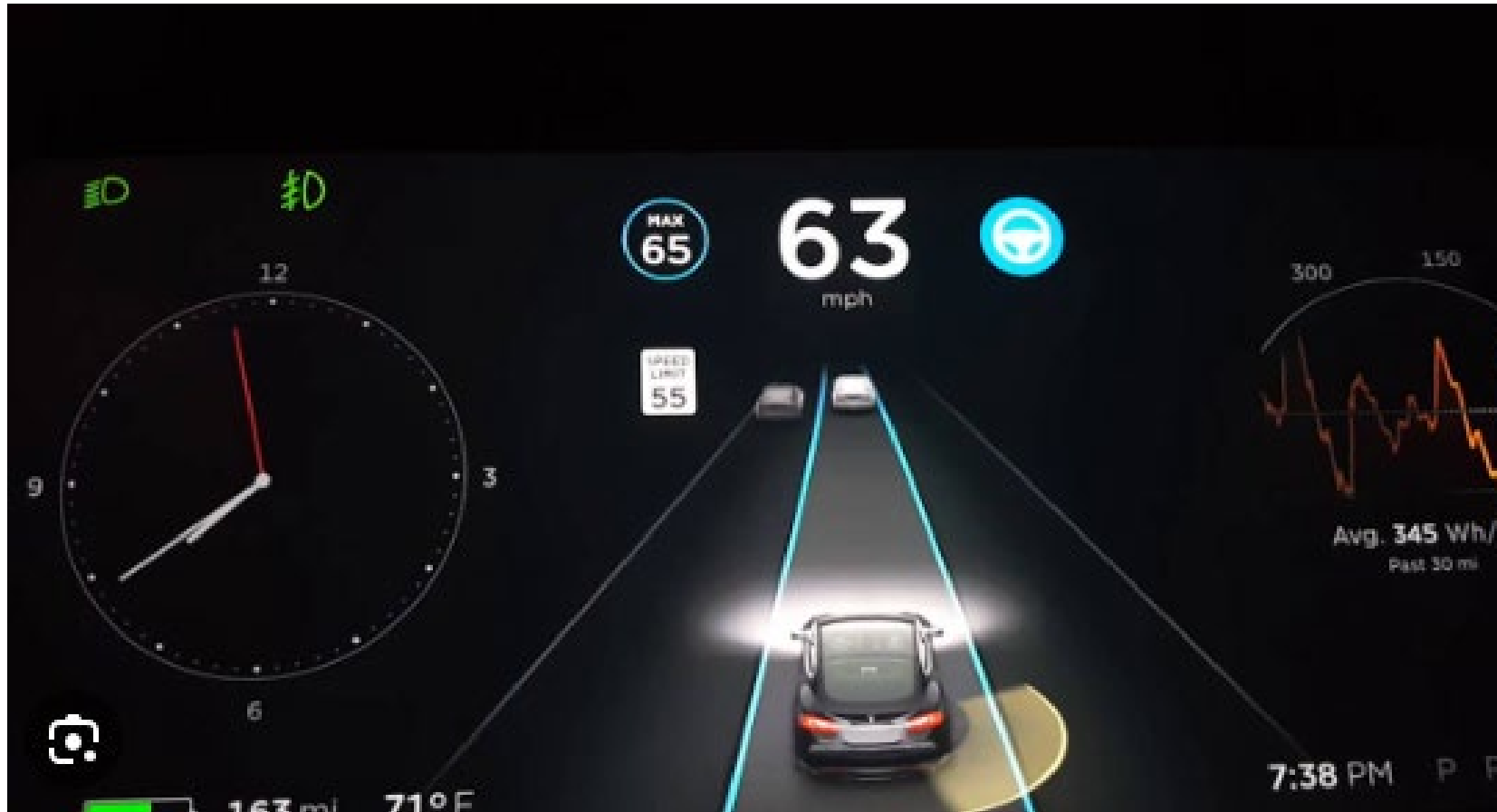


A human driver has guidance control of the vehicle under normal conditions.

Suppose the driver sees an animal, but does not change lanes to avoid it.

Counterfactually, had she tried to change lanes, the autopilot would have blocked her.

Is the Tesla driver responsible for the animal's death, even though automation made sure that she couldn't do otherwise?



Reverse case: Autopilot has guidance control of the vehicle under normal conditions.

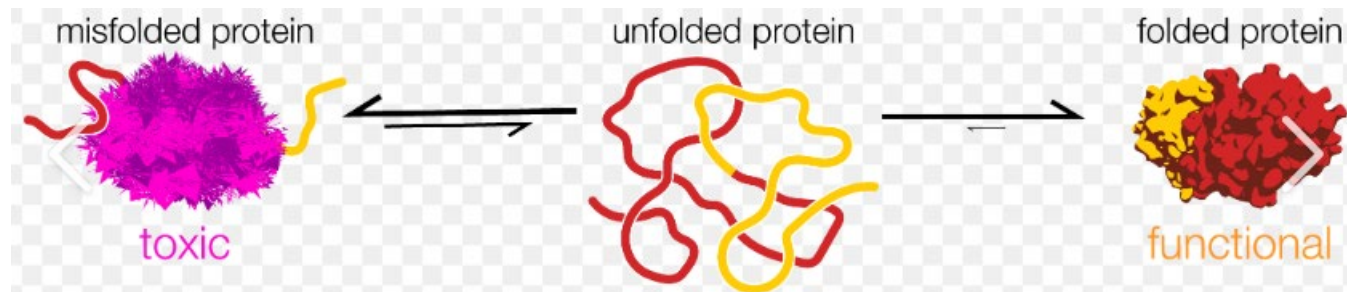
Suppose as a foreseeable consequence of its design, autopilot does not change lanes to avoid an animal.

Suppose that had Autopilot tried to change lanes, the human operator would have blocked it.

Are the designers of Autopilot responsible for the animal's death, even though the human operator made sure that it couldn't do otherwise?

Zooming out: when should we automate?

- “While AlphaFold can assume significant tasks previously done by human scientists (i.e., determining protein structures) this should positively impact, or at least have a neutral effect, on task integrity if it allows scientists to re-focus their work efforts on other important aspects of their broader goal of curing diseases. However, there remain risks to AI being used in this way. Continuing with this example, if scientists have trained for many years to do the experimental work that AlphaFold can now do more quickly and accurately, this generates significant risks for their ability to exercise their full capacities, demonstrate their mastery, and utilise the skills they have invested years in developing to reach their full potential” (Bankins & Formosa 2023).



The claims of MHC and related value paradigms

High levels of automation can be combined with human contributions, in a way that allows us to achieve the following values in AI and robotics:

- Responsibility (MHC)
- Trustworthiness (Shneiderman 2019)
- Amplification of human work (Bankins & Formosa 2023)
- “Hybrid Intelligence”: Synergy of human and machine intelligence (Akata et al. 2020)

Automating aspects of *judgment*

- Suppose a person's discretion in judgment (e.g., professional, aesthetic, scientific, or moral judgment) is important to their authority and responsibility
- Now consider automation of “judgment-supporting” tasks:
 - Deciding on the parameters of judgment (what scale to use)
 - Filtering the evidence and deciding when one has enough evidence
 - Taking a second look/ checking one's work
 - Noting consistencies or inconsistencies across time and context
 - Providing an account or explanation of the judgment after the fact
- Here the human retains control. But human freedom can be reduced (depending how the social situation constitutes it), along with trust and responsibility.

The risks of de-skilling

- ***Deskilling*** refers to the gradual erosion of knowledge, judgment, and know-how through overreliance on others, cultural loss, or overreliance on automation. By extension, it also refers to the prevention of learning through these mechanisms.
- ***Reliance on intelligent automation has proven effects of deskilling*** among professionals (Heudel et al. 2026).

“Evidence of clinical deskilling, though scarce, is consistent across specialties” (Heudel et al. 2026)

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